BAS-301	Mathematics III	L T P 3 1 0	4 credits

Course Objectives:

CO1	To understand the method of solving algebraic, transcendental equations.
CO2	determine the approximate value of the derivative & definite integral for a given data using numerical techniques.
CO3	to expand the given periodic function defined in the given range in terms of sine and cosine multiple of terms as a Fourier series.
CO4	to extreme the functional using integration technique.
CO5	know how root finding techniques can be used to solve practical engineering problems.

MODULE-I

Differential Equations

Linear differential equations of nth order with constant coefficients, Complementary functions and particular integrals, Simultaneous linear differential equations, Solutions of second order differential equations by changing dependent and independent variables, Method of variation of parameters, Applications to engineering problems (without derivation).

Series Solutions and Special Functions Series solutions of ODE of 2nd order with variable coefficients with special emphasis to differential equations of Legendre, and Bessel Legendre polynomials, Bessels functions and their properties.

MODULE-II

Partial Differential Equations

Introduction of partial differential equations, Linear partial differential equations with constant coefficients of 2nd order and their classifications - parabolic, elliptic and hyperbolic with illustrative examples.

Applications of Partial Differential Equations

Method of separation of variables for solving partial differential equations, Wave and Heat equations in one dimension. Laplace equation.

MODULE-III

Laplace Transform

Laplace transform, Existence theorem, Laplace transform of derivatives and integrals, Inverse Laplace transform, Unit step function. Laplace transform of periodic functions, Convolution theorem, Application to solve simple linear and simultaneous differential equations.

Function of Complex variable

Analytic function, C-R equations, Cauchy's integral theorem, Cauchy's integral formula for derivatives of analytic function, Laurent's series, singularities, Residue theorem, Evaluation of real integrals of the type $\int_0^{2\pi} f(Cos\,\theta, Sin\,\theta)\,d\theta$ and $\int_{-\infty}^{\infty} f(x)\,dx$

Text Books:

1. H.K.Dass, Higher Engineering Mathematics, S.Chand Publications.
2. B.S.Grewal, Higher Engineering Mathematics, Khanna Publishers, 2005

Reference Books:-

1. R.K.Jain & S.R.K.Iyenger, Advance Engineering Mathematics, Narosa Publishing House, 2002.
2. E.Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, 2005.
3. C.Ray Wylie & Louis C. Barrett, Advanced Engineering Mathematics, Tata Mc Graw-Hill Publishing Company Ltd. 2003
4. Peter V. O'Neil, Advanced Engineering Mathematics, Thomson (Cengage) Learning, 2007.

COURSE OUTCOMES:

CO1	Apply the Set theory and Relation concepts
CO2	Apply the Functions and define the recursive functions.
CO3	Apply Laplace transform to different applications
CO4	Apply Inverse Laplace transform to different applications.
CO5	Identify the permutations and combinations.